

Risk-Adjusted Portfolio Benchmarking via Latent Performance Decomposition

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Abstract

Assessing portfolio performance relative to systematic risk factors is a central task in empirical asset pricing and quantitative investment analysis. Conventional approaches typically rely on regression-based alphas or average excess returns, which treat deviations from benchmark returns as symmetric noise and do not explicitly distinguish between transitory variation and persistent shortfall relative to an achievable benchmark.

This project develops a latent performance decomposition framework for benchmarking equity portfolios against standard risk factors. Portfolio returns are modelled relative to a factor-implied performance envelope, allowing realised outcomes to be decomposed into stochastic fluctuations and a latent component capturing systematic underperformance. This latent component is used to construct a risk-adjusted efficiency measure that is comparable across portfolios and robust to noise.

The framework is applied to long-horizon equity portfolios formed on size and book-to-market characteristics using established factor benchmarks. Both static cross-sectional estimates and rolling-window analyses are conducted to assess the persistence and stability of latent performance measures over time. The empirical results demonstrate that latent performance rankings exhibit strong persistence and structured mobility, providing a complementary perspective on portfolio benchmarking that extends beyond conventional alpha-based metrics.

1 Background and Motivation

Evaluating portfolio performance in a risk-adjusted setting is a central problem in finance. Since the development of multifactor asset pricing models, empirical performance assessment has largely focused on estimating abnormal returns relative to factor benchmarks. In this framework, regression intercepts and residual-based statistics are commonly interpreted as measures of outperformance or underperformance after controlling for systematic risk exposures.

This regression-based approach has become standard in both academic research and applied quantitative analysis. Factor models are widely used to explain cross-sectional variation in returns, attribute portfolio performance, and benchmark active investment strategies. However, conventional regression-based evaluation implicitly treats deviations from benchmark returns as symmetric noise. As a result, it does not explicitly distinguish between random, transitory

variation and persistent shortfall relative to the level of performance implied by a portfolio's risk characteristics.

Several strands of the literature have highlighted limitations of interpreting regression residuals or estimated alphas as comprehensive measures of performance. Abnormal return estimates can be sensitive to sample length, factor specification, and noise, particularly in long-horizon settings. Moreover, portfolios that systematically fail to achieve returns consistent with their factor exposures may appear statistically similar to portfolios whose deviations are largely driven by idiosyncratic fluctuations, obscuring meaningful differences in underlying performance quality.

In parallel, a broader body of work in econometrics and productivity analysis has developed methods for separating random noise from systematic inefficiency relative to an attainable benchmark. These approaches emphasise envelope or frontier representations, in which observed outcomes are evaluated relative to a performance boundary rather than a central tendency. While such methods are well established in non-financial settings, their application to portfolio performance benchmarking has been comparatively limited.

This project builds on these insights by developing a latent performance decomposition framework for risk-adjusted portfolio benchmarking. Rather than focusing solely on average abnormal returns, the approach models an upper performance envelope implied by observed factor exposures and decomposes realised returns into two components: a stochastic disturbance capturing random variation and a latent component representing systematic shortfall relative to the benchmark. This latent component provides a measure of risk-adjusted efficiency that captures persistent underperformance while remaining robust to transitory noise.

The framework is applied to equity portfolios formed on size and book-to-market characteristics and benchmarked against standard factor models. Both full-sample cross-sectional estimates and rolling-window analyses are conducted to examine the stability and persistence of latent performance measures over time. By integrating ideas from factor-based asset pricing and latent performance modelling, the analysis offers a complementary perspective on portfolio benchmarking that extends beyond conventional regression-based metrics.

The emphasis of this work is practical benchmarking and interpretability rather than formal hypothesis testing or publication-oriented inference. This document is intended as a technical demonstration of a portfolio benchmarking framework, focusing on methodological clarity, empirical behaviour, and relevance for applied quantitative analysis and model development.

2 Methodology

This section presents the latent performance decomposition framework used to benchmark portfolio performance relative to risk factors. The objective is to separate random variation in returns from systematic shortfall relative to a factor-implied benchmark, and to construct a risk-adjusted efficiency measure that is comparable across portfolios and robust to transitory noise.

Let $R_{p,t}$ denote the return of portfolio p in period t , and let $R_{f,t}$ denote the risk-free rate. Excess returns are defined as

$$r_{p,t} = R_{p,t} - R_{f,t}. \quad (1)$$

Portfolio excess returns are benchmarked against a linear factor model of the form

$$r_{p,t} = \beta_{p,0} + \boldsymbol{\beta}_p^\top \mathbf{f}_t + \varepsilon_{p,t}, \quad (2)$$

where $\mathbf{f}_t$ is a vector of observed risk factors and $\boldsymbol{\beta}_p$ denotes the corresponding factor loadings. In the empirical application, the factor vector includes the market excess return, size, and value factors.

In conventional regression-based evaluation, deviations from this specification are treated as symmetric noise. In contrast, the framework developed here allows for an asymmetric decomposition of residuals that distinguishes between random fluctuations and systematic shortfall relative to the factor-implied benchmark. Specifically, observed deviations are decomposed as

$$\varepsilon_{p,t} = v_{p,t} - u_{p,t}, \quad (3)$$

where $v_{p,t}$ is a symmetric stochastic disturbance capturing transitory variation, and $u_{p,t} \geq 0$ is a non-negative latent component representing systematic underperformance relative to the benchmark.

The stochastic disturbance $v_{p,t}$ is assumed to have zero mean and to capture idiosyncratic shocks, measurement error, and other sources of random variation. The latent component $u_{p,t}$ captures persistent shortfall relative to the achievable performance implied by the portfolio's factor exposures. By construction, this component shifts realised returns downward from the benchmark and is not permitted to take negative values.

This asymmetric decomposition induces a frontier interpretation of benchmark performance. Conditional on factor exposures, the factor model defines an upper envelope of attainable returns, while realised outcomes may fall below this envelope due to latent inefficiencies or structural frictions. Portfolios are therefore evaluated relative to a performance boundary rather than a conditional mean.

The latent shortfall component is mapped into a unit-free risk-adjusted efficiency measure via the transformation

$$\text{AE}_{p,t} = \exp(-u_{p,t}), \quad (4)$$

where $\text{AE}_{p,t} \in (0, 1]$. Values close to one indicate that realised returns are close to the factor-implied benchmark, while lower values indicate increasing degrees of systematic underperformance. Because the measure is derived from the latent component rather than raw residuals, it is less sensitive to transitory noise and directly comparable across portfolios with different risk characteristics.

Portfolio-level efficiency summaries are constructed by averaging $\text{AE}_{p,t}$ over time, yielding cross-sectional rankings based on persistent, risk-adjusted performance rather than average abnormal returns.

To assess stability and persistence, the decomposition is estimated both over the full sample and within rolling time windows. Rolling-window estimation allows efficiency measures to evolve over time while maintaining a consistent benchmark structure. For a given window length W , the model is re-estimated over overlapping windows $\{t - W + 1, \dots, t\}$, producing time-varying efficiency estimates that are used to study rank persistence, mobility, and transitions across

performance groups.

The proposed framework is not intended to identify abnormal returns in the traditional sense, nor to assert market inefficiency. Instead, it provides a practical benchmarking tool that decomposes realised performance into random noise and systematic shortfall relative to a factor-based envelope. By focusing on latent underperformance rather than average residuals, the approach offers a complementary perspective that is particularly well suited to long-horizon analysis and comparative evaluation across portfolios.

3 Data

The empirical analysis uses equity portfolio returns constructed on firm size and book-to-market characteristics. Specifically, 25 portfolios are formed at the intersection of five size groups and five book-to-market groups, following standard portfolio construction conventions widely used in factor-based asset pricing. Portfolio assignments are updated annually, and value-weighted monthly returns are computed over the subsequent year.

The portfolio return series span a long historical sample beginning in July 1926 and extending through November 2025. Returns are observed at a monthly frequency and are expressed in percentage units in the original data. For the purposes of analysis, all returns are converted to decimal form and aligned to month-end timestamps. Portfolios with missing return observations are excluded on a month-by-month basis, and no interpolation is performed, ensuring that all estimates are based exclusively on observed data.

Risk adjustment is conducted using a standard set of equity risk factors. The factor set includes the market excess return, a size factor, and a value factor, along with the one-month risk-free rate. All factor series are observed at a monthly frequency and are aligned to the same calendar dates as the portfolio returns. The risk-free rate is subtracted from portfolio returns to construct excess returns prior to estimation. As with the portfolio returns, factor series are converted from percentage units to decimals, and observations flagged as missing using sentinel values in the source files are removed prior to analysis.

Portfolio returns and factor data are merged by calendar month using an inner join, ensuring that only periods with complete information are included. The resulting dataset contains one observation per portfolio per month and includes excess returns and factor values required for estimation. To ensure stability of parameter estimates, portfolios are required to have a minimum number of observations before being included in the analysis. Owing to the long historical sample, all portfolios in the final dataset exceed this threshold by a substantial margin.

In addition to the full-sample dataset, rolling-window datasets are constructed to analyse time variation in latent performance measures. For each portfolio, overlapping estimation windows of fixed length are formed, and the model is re-estimated sequentially through time. This rolling structure preserves the original data frequency and alignment while allowing risk-adjusted performance measures to evolve dynamically. The resulting panel forms the basis for persistence, transition, and mobility analysis in subsequent sections.

Summary statistics for portfolio returns and factor variables are reported in the Appendix. These statistics provide context for the magnitude and variability of returns over the sample

Table 1: Static Cross-Sectional Performance Rankings

Portfolio	Mean AE	Rank
ME4 BM3	0.867	1
ME4 BM1	0.861	2
ME4 BM4	0.860	3
ME3 BM3	0.855	4
ME4 BM2	0.855	5
ME3 BM2	0.854	6
ME4 BM5	0.854	7
ME3 BM1	0.853	8
ME3 BM4	0.852	9
ME1 BM3	0.837	10
ME1 BM2	0.836	11
ME3 BM5	0.834	12
ME1 BM4	0.816	13
SMALL HiBM	0.801	14
SMALL LoBM	0.783	15
ME2 BM1	0.778	16
ME2 BM2	0.773	17
ME2 BM3	0.772	18
ME2 BM4	0.768	19
ME2 BM5	0.750	20
BIG LoBM	0.704	21
ME5 BM2	0.696	22
ME5 BM4	0.695	23
BIG HiBM	0.688	24
ME5 BM3	0.672	25

period and confirm that the data are consistent with well-established stylised facts in the equity returns literature.

4 Empirical Results

This section presents empirical results from applying the latent performance benchmarking framework to size–value equity portfolios. We first examine full-sample cross-sectional rankings, followed by an analysis of temporal stability using rolling-window estimates. Transition dynamics and mobility metrics are then used to characterise the persistence and evolution of latent performance over time.

4.1 Static Cross-Sectional Performance

Full-sample estimates reveal clear and economically meaningful cross-sectional differences in latent, risk-adjusted performance across portfolios. Portfolios formed on intermediate size and book-to-market characteristics consistently achieve the highest efficiency scores, while large-cap portfolios—particularly those at the extremes of the value dimension—tend to exhibit lower latent efficiency.

Because efficiency scores are averaged over a long historical sample, these rankings reflect persistent performance characteristics rather than short-term fluctuations. Adjacent portfolios display smooth variation in efficiency, with no evidence of erratic rank reversals or instability, suggesting that the latent performance measure captures a coherent cross-sectional structure.

Importantly, the resulting rankings do not mechanically coincide with raw average returns or factor loadings. Instead, they reflect systematic shortfall relative to factor-implied benchmarks, highlighting differences in how reliably portfolios realise the performance implied by their risk exposures. This distinction illustrates the value of the latent decomposition relative to conventional alpha-based summaries.

4.2 Temporal Stability and Rank Persistence

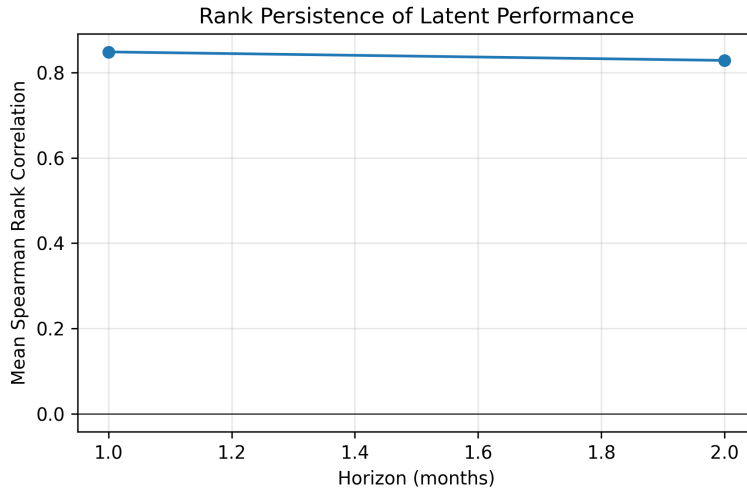


Figure 1: Mean Spearman rank correlation of latent performance rankings across rolling windows. High short-horizon persistence and only modest decay at longer horizons indicate that latent performance captures stable cross-sectional characteristics rather than transitory noise.

Figure 1 summarises the temporal stability of latent performance rankings using rolling-window estimates. The mean Spearman rank correlation between consecutive windows is approximately 0.85 at a one-month horizon and remains high at longer horizons, exhibiting only modest decay.

This degree of persistence is inconsistent with noise-driven ranking variation and indicates that latent performance captures durable cross-sectional differences across portfolios. The results suggest that portfolios tend to maintain their relative efficiency ordering over time, supporting the interpretation of latent performance as a structural characteristic rather than a transient signal.

4.3 Quintile Transition Dynamics

Additional insight into ranking dynamics is provided by the quintile transition matrix in Figure 2. Transition probabilities are heavily concentrated along the diagonal, particularly for the top and bottom quintiles, indicating substantial persistence in relative performance.

Table 2: Quintile Transition Probabilities

From	1.0	2.0	3.0	4.0	5.0
1	0.875	0.064	0.021	0.016	0.023
2	0.068	0.688	0.176	0.034	0.034
3	0.020	0.179	0.620	0.159	0.022
4	0.018	0.031	0.158	0.640	0.153
5	0.019	0.039	0.024	0.152	0.767

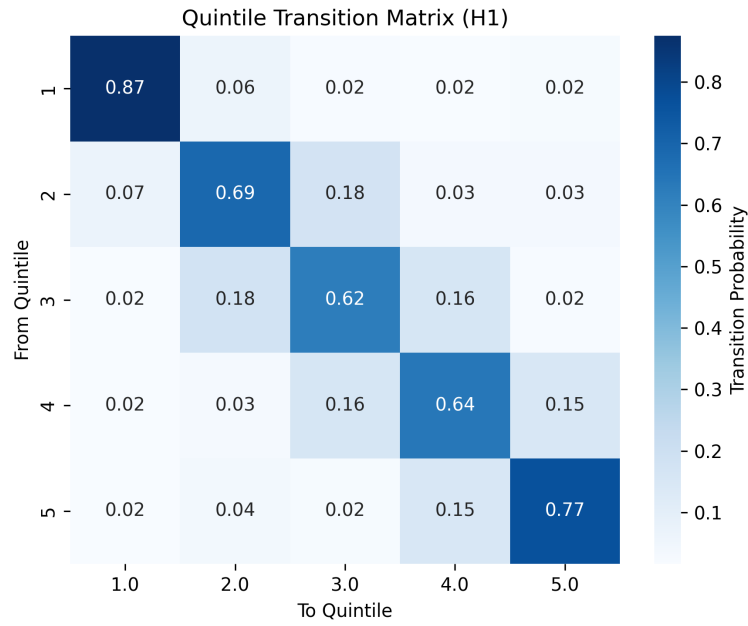


Figure 2: Quintile transition matrix for latent performance rankings. Transition probabilities are strongly concentrated along the diagonal, indicating substantial persistence. Movements are predominantly between adjacent quintiles, with large jumps being rare.

Table 3: Quintile Mobility Metrics

Quintile	Stay	Improve	Deteriorate
1	0.875	0.000	0.125
2	0.688	0.068	0.244
3	0.620	0.198	0.181
4	0.640	0.208	0.153
5	0.767	0.233	0.000

When transitions do occur, they are predominantly between adjacent quintiles. Large jumps across multiple quintiles are rare, suggesting that changes in latent performance tend to be gradual rather than abrupt. This behaviour is consistent with the interpretation of latent efficiency as a slowly evolving performance attribute rather than a volatile ranking metric.

4.4 Mobility Patterns

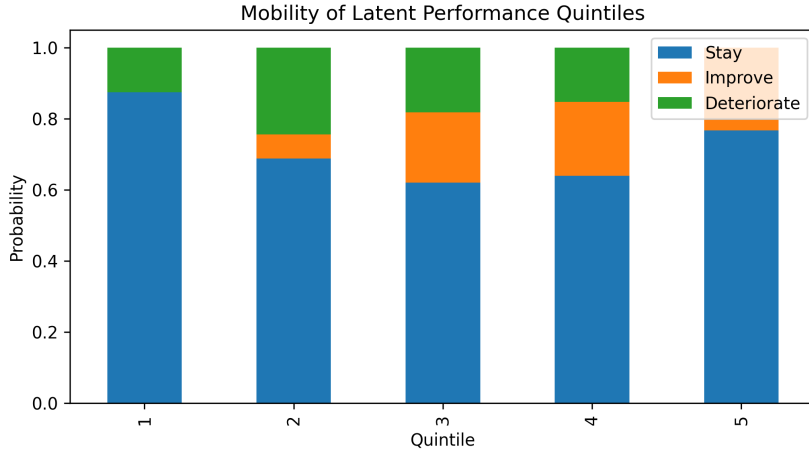


Figure 3: Mobility decomposition of latent performance rankings. Bars show the probability of remaining in the same quintile, improving, or deteriorating between consecutive rolling windows. Extreme quintiles exhibit higher stability, while middle quintiles display more balanced upward and downward mobility.

Figure 3 decomposes quintile transitions into probabilities of staying, improving, or deteriorating between consecutive windows. Extreme quintiles exhibit the highest stability, with portfolios in the top quintile remaining in place nearly 88% of the time and portfolios in the bottom quintile exhibiting similarly strong persistence.

In contrast, middle quintiles display more balanced upward and downward mobility, reflecting greater scope for relative movement. Notably, downward movement from the top quintile is less frequent than upward movement from the bottom quintile, indicating asymmetric mobility consistent with persistent performance advantages among high-efficiency portfolios.

Overall, the mobility patterns reinforce the view that latent performance rankings reflect stable structural differences across portfolios, with limited short-term reordering and gradual evolution over time.

5 Conclusion

This technical note developed a framework for *risk-adjusted portfolio benchmarking via latent performance decomposition*. Rather than relying solely on average abnormal returns or regression residuals, the approach decomposes realised portfolio returns into a stochastic component and a latent performance shortfall relative to a factor-implied upper envelope. The resulting latent performance measure provides a stable and interpretable indicator of relative portfolio efficiency after controlling for systematic risk exposures.

Applied to size- and book-to-market–sorted equity portfolios, the empirical analysis demonstrates several robust properties. First, latent performance rankings exhibit clear and economically meaningful cross-sectional structure and are not dominated by transitory noise. Second, rolling-window analysis reveals a high degree of rank persistence, with only modest decay over longer horizons. Third, transition and mobility analyses show that extreme performance groups are particularly stable, while middle-ranked portfolios exhibit more balanced upward and downward movement. These patterns are difficult to reconcile with purely stochastic deviations and suggest that latent performance captures persistent differences in risk-adjusted portfolio behaviour.

From a practical perspective, the framework offers a complementary alternative to conventional factor-based performance evaluation. By focusing on persistent shortfall relative to achievable benchmarks, it enables more nuanced diagnostics of portfolio behaviour, including stability, deterioration risk, and improvement potential. The methodology is fully data-driven, extensible to alternative factor specifications, and naturally suited to rolling or real-time monitoring environments.

Overall, the results indicate that latent performance decomposition provides a useful and informative lens for portfolio benchmarking and risk-adjusted performance analysis. The framework is well suited for quantitative research and applied portfolio diagnostics, and it offers a flexible foundation for further extensions in systematic strategy evaluation and performance monitoring.